

Anomalous fractal scaling in two-dimensional electric networks



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Reviewers' comments:

Reviewer #1 (Remarks to the Author):

The authors theoretically investigated and experimentally measured the electrical impedance in 2D (R)LC circuit networks.

They showed that the complex impedance circuits could resonate under certain conditions, distinct from the resistor circuits.

They also reported the deviation from logarithmic scaling in the impedance, which, in their view, merely occurs in 2D LC circuit lattices. This anomalous behavior is traced back to the resonance condition obtained from network Laplacian. Using the resonance condition for the impedance, the authors determined the resonance frequency at which they observed their anomalous structure in impedance plots. Finally, they presented that the resonance peaks are robust against perturbations from parasite contributions.

The applicability extent of electrical circuits as classical platforms to explore various topological and non-Hermitian models is growing. It is thus essential to have a better understanding of these circuits. For this reason, the attempt of authors to shed light on the distinct properties of active circuits can be helpful. Nevertheless, several points prevent the recommendation of the current manuscript for publication. After addressing the following points, I will judge whether the article deserves publication in Communications Physics.

Here are the major comments.

1) In the third paragraph, it is written, " we report the discovery of a dramatic logarithmic scaling violation in the impedance across 2D LC circuit lattices". However, what they call their discovery has been known for a while. In fact, the resonance of complex impedance in (R)LC circuits has been previously studied. Here are some of the relevant papers:

- RLC network:

Zhi-Zhong Tan, "Two-point resistance of an $m \times n$ resistor network with an arbitrary boundary and its application in RLC network," Chinese Phys. B 25 050504 (2016).

-RLC network:

Zhou, L., Tan, Zz. and Zhang, Qh. "A fractional-order multifunctional n-step honeycomb RLC circuit network," Frontiers Inf Technol Electronic Eng 18, 1186–1196 (2017).

-LC network:

Tan, Z.-Z., Asad, J. H., and Owaidat, M. Q. "Resistance formulae of a multipurpose n-step network and its application in LC network, Int. J. Circ. Theor. Appl., 45: 1942– 1957 (2017).

-2xn LC network:

Zhen Tan et al., "Electrical characteristics of the $2 \times n$ and $\Box \times n$ circuit network", Phys. Scr. 94 055203 (2019).

-LC ladder network:

Hai-Xiang Chen, Lei Yang, Ming-Jun Wang,
"Electrical characteristics of n-ladder network with internal load," Results in Physics, Volume 15, 102488 (2019).

- LC triangular network:

Hai-Xiang Chen, Na Li, Zi-Tian Li, Zhi-Zhong Tan,
"Electrical characteristics of a class of n-order triangular network," Physica A: Statistical Mechanics

and its Applications,
Volume 540, 123167 (2020).

-N-order LC network with Y circuits:

Hai-Xiang Chen and Zhi-Zhong Tan, "Electrical properties of an n-order network with Y circuits," Phys. Scr. 95 085204 (2020).

-LC network:

Chen, HX., Yang, L. "Electrical characteristics of n-ladder network with external load," Indian J Phys 94, 801–809 (2020).

-LC network:

Cui-Ping Chen, Zhi-Zhong Tan, "Electrical characteristics of an asymmetric N-step network," Results in Physics, Volume 19, 103399,(2020).

-Two-layer 2D RLC circuits:

Ammar, N, Asad, J, Jarrar, R. "Electrical characteristics for triangular resistors–capacitors–inductors network designed on two coats," Int J Circ Theor Appl., 50(1): 153- 174 (2022).

-LC network:

Fang, XY., Tan, ZZ. "Circuit network theory of n-horizontal bridge structure," Sci Rep 12, 6158 (2022).

Despite this, nearly all of these papers are theoretical; hence, reporting the experimental measurement of oscillating impedance in this work is important.

2) To motivate their study, the authors present a one-parameter scaling theory within which the anomalous impedance scales as $Z \propto N^{2-D}$ for $D \neq 2$ and $Z \propto \log N$ when $D=2$. While this is an intuitive picture, I am concerned about its validity and reliability. This is because, for any (R)LC circuit, independent of dimension, one can evaluate the Laplacian and, subsequently, determine the resonance frequencies responsible for irregular oscillations in the imaginary parts of impedance.

Hence, independent of the dimension of these resonant circuits, one should be able to determine their resonance frequencies and observe anomalous impedance. Here are references that reported similar structure in the impedance in quasi-1D (1xn) ladder LC network and n-step 3D cube LC network.

-Ladder LC network:

Hai-Xiang Chen et al., "Equivalent complex impedance of n-order RLC network," Phys. Scr. 96 075202 (2021)

-n-step cube network:

Zhen Tan et al., "Electrical characteristics of the $2 \times n$ and $\sqrt{2} \times n$ circuit network" Phys. Scr. 94 055203 (2019).

The reported oscillating impedance in both papers is in dimensions different than two. This undermines the authors' argument regarding the importance of " $D=2$ " and the connection they drew between their results and other 2D systems in other research fields.

Here are other comments and questions.

-The manuscript should include a proper introduction to previous works on complex impedance

networks.

-For better readability of this work in the physics community and to keep the manuscript self-contained, it would be helpful if the authors present the relation between impedance, resistance, and reactance, either as a footnote or in the text.

-As Z is imaginary in LC circuits, one should use $|Z|$ instead of Z in the definition of real-valued β , plotted in Fig.1.

-For the benefit of readers, the recursive relation of the circuit should be calculated and added in the main text, which can then be directly compared with Harper's equation.

-The authors described the anomalous structure of impedance as fractal-like. Could they determine the fractal dimension or the rescaling relation?

For resistance networks, a simple rescaling ratio, not fractal though, in certain geometries is introduced here: Rajat Chandra Mishra and Himadri Barman, "Effective resistances of two-dimensional resistor networks" Eur. J. Phys. 42 015205 (2020).

-The authors also discussed the fractal parameter space profile of the impedance. In disordered conductors, one may encounter multifractality; see, for instance, Phys. Rev. Lett. 94, 244102 (2005) and Phys. Rev. E 73, 036210 (2006). In their study, could authors draw or rule out a link between multifractality in conductors and their reported fractal structures?

Reviewer #2 (Remarks to the Author):

In this manuscript, the authors find an anomalous impedance with pronounced and erratically located peaks in a 2D LC lattice, which is different from the logarithmic impedance scaling in the continuum media. Such anomalous fractal scaling of impedance is confirmed in experiments, and attributed to the circuit resonance, which is related with commensurability conditions for the vanishing of circuit Laplacian eigenvalues. They further show that such impedance peaks are robust against the contact and parasitic resistances. I think this research is timely and interesting and, to the best of my knowledge, scientifically sound, thus recommend it for publication in Communications Physics.

I would like the authors to address the following comments:

(1) Why do the impedance peaks in experiments shift to right regularly? It may be not because of the uncertainly tolerances of the electronic elements. Maybe it is owing to the global loss induced by the inevitable resistances. I suggest the authors can give a more detail discussion about this point.

(2) Why is the range of w_r in Fig. 3(a) and 3(c) different? Also the color used to denote the value of $\log(Z)$? The same range and color would be more easily to compare these two results, revealing the effect of the inevitable resistances.

(3) Although the authors write a sentence to discuss the case of generic circuit, I hope they can discuss the case of honeycomb lattice of LC with 2 site in a unit cell in detail, which should be an important extension of this work.

(4) I suggest the authors can optimize the distribution of the figures, to make them look clear. For example, six figures in a row in Fig. 2 result in each figure small. And the labels of figures, (a),(b),(c)...., are better to locate at the left top of the figures.

Reviewer #3 (Remarks to the Author):

In their paper, Xiao Zhang and coauthors consider a circuit in the form of the square lattice with horizontal and vertical links between adjacent nodes being capacitors and inductance coils, respectively. The authors reveal counterintuitive dependence of the corner-to-corner impedance on the number of nodes in the circuit, which demonstrates the emergence of pronounced peaks for certain numbers of nodes (considering some given frequency of the excitation) instead of a logarithmic scaling.

In my opinion, the paper needs to be revised prior publication.

First, the authors include a lot of references to papers considering topological electrical circuits (most of them being self-citations) many of which can be omitted as the present paper is not related to topological effects, while not including any citation for the considered circuit design. The studied circuit is not novel and has been considered, for example, in [Z. Tan, Chin. Phys. B 25, 050504 (2016)], where the expressions for impedance between two arbitrary nodes at the same side of the circuit were obtained. This reference should be included in the manuscript, and I would like to recommend the authors to check carefully if the considered circuit has been studied in any other papers. Also, slightly modified circuits were considered to implement hyperbolic metamaterials [P. Kapitanova, et al., Nat. Commun. 5, 3226 (2014)].

I think that the novelty of the present manuscript is not in the circuit design, but in revealing commensurability properties and establishing a connection between such a circuit and properties of electrons in a magnetic field, which seem to be a very interesting result. In the light of such an analogy, I would also like to attract the attention of the authors to the paper [A.V. Poshakinskiy, et al., npj Quantum Inf. 7, 34 (2021)] considering a correspondence between Aubry-Andre-Harper model and the system of coupled microwave qubits located in a waveguide.

Regarding the discretization of continuous media and its representation with electrical circuits, I would like the authors to consider mentioning well known papers [S. Kirkpatrick, Rev. Mod. Phys. 45, 574 (1973); I. Webman, et al., Phys. Rev. B 15, 5712 (1977)]. Besides, some references on more general networks should be introduced in Discussion to illustrate the statements written in this section.

Also, there are several minor concerns that should be addressed in order to improve the general quality of the paper.

1. There are several typos and mistakes in English throughout the text.
2. Many of the references contain various imperfections in formatting.
3. In Introduction and other sections of the manuscript, LC circuits should be mentioned as a resonant media instead of just a reactive one, as purely capacitive or inductive circuits are reactive as well. However, they obviously will not demonstrate any unusual behavior on the number of nodes in the circuit.

The paper is clearly organized and well written.

Response to Reviewers' Comments/Questions

Response to Reviewer #1:

Main comment

The authors theoretically investigated and experimentally measured the electrical impedance in 2D (R)LC circuit networks. They showed that the complex impedance circuits could resonate under certain conditions, distinct from the resistor circuits.

They also reported the deviation from logarithmic scaling in the impedance, which, in their view, merely occurs in 2D LC circuit lattices. This anomalous behavior is traced back to the resonance condition obtained from network Laplacian. Using the resonance condition for the impedance, the authors determined the resonance frequency at which they observed their anomalous structure in impedance plots. Finally, they presented that the resonance peaks are robust against perturbations from parasite contributions.

The applicability extent of electrical circuits as classical platforms to explore various topological and non-Hermitian models is growing. It is thus essential to have a better understanding of these circuits. For this reason, the attempt of authors to shed light on the distinct properties of active circuits can be helpful. Nevertheless, several points prevent the recommendation of the current manuscript for publication. After addressing the following points, I will judge whether the article deserves publication in Communications Physics.

Response: We appreciate the detailed review and valuable comments provided by the reviewer. We believe that our manuscript offers valuable contributions to the field by comprehensively revealing the size-dependent resonances and their fractal-like nature in the size and parameter domain. As the reviewer rightly pointed out, classical electrical circuits are increasingly being employed to realize tight-binding lattice models, such as topoelectrical circuits and non-Hermitian phenomena, due to their straightforward tunability of lattice hoppings. Consequently, understanding the lattice dynamics, including impedance resonances and their underlying conditions in such electrical circuits, has become essential.

In our manuscript, we study the resonance dynamics of LC circuits and present the anomalous size-dependent impedance resonances, which have not been comprehensively presented before, and trace back their origins, as well as their experimental demonstration via a 2D LC circuit. Therefore, we believe our results shed light on the generic conditions of LC lattice models.

We acknowledge that there are several points that need to be addressed before our manuscript can be recommended for publication in Communications Physics. Based on the concerns and questions raised, we have carefully revised our manuscript and addressed each point below.

Comments and questions

- 1) In the third paragraph, it is written, " we report the discovery of a dramatic logarithmic scaling violation in the impedance across 2D LC circuit lattices". However, what they call

their discovery has been known for a while. In fact, the resonance of complex impedance in (R)LC circuits has been previously studied. Here are some of the relevant papers:

- a) **[R1-1]** RLC network:
Zhi-Zhong Tan, "Two-point resistance of an $m \times n$ resistor network with an arbitrary boundary and its application in RLC network," Chinese Phys. B 25 050504 (2016).
- b) **[R1-2]** RLC network:
Zhou, L., Tan, Zz. and Zhang, Qh. "A fractional-order multifunctional n-step honeycomb RLC circuit network," Frontiers Inf Technol Electronic Eng 18, 1186–1196 (2017).
- c) **[R1-3]** LC network:
Tan, Z.-Z., Asad, J. H., and Owaidat, M. Q. "Resistance formulae of a multipurpose n-step network and its application in LC network, Int. J. Circ. Theor. Appl., 45: 1942– 1957 (2017).
- d) **[R1-4]** $2 \times n$ LC network:
Zhen Tan et al., "Electrical characteristics of the $2 \times n$ and $m \times n$ circuit network", Phys. Scr. 94 055203 (2019).
- e) **[R1-5]** LC ladder network:
Hai-Xiang Chen, Lei Yang, Ming-Jun Wang, "Electrical characteristics of n-ladder network with internal load," Results in Physics, Volume 15, 102488 (2019).
- f) **[R1-6]** LC triangular network:
Hai-Xiang Chen, Na Li, Zi-Tian Li, Zhi-Zhong Tan, "Electrical characteristics of a class of n-order triangular network," Physica A: Statistical Mechanics and its Applications, Volume 540, 123167 (2020).
- g) **[R1-7]** N-order LC network with Y circuits:
Hai-Xiang Chen and Zhi-Zhong Tan, "Electrical properties of an n-order network with Y circuits," Phys. Scr. 95 085204 (2020).
- h) **[R1-8]** LC network:
Chen, HX., Yang, L. "Electrical characteristics of n-ladder network with external load," Indian J Phys 94, 801–809 (2020).
- i) **[R1-9]** LC network:
Cui-Ping Chen, Zhi-Zhong Tan, "Electrical characteristics of an asymmetric N-step network," Results in Physics, Volume 19, 103399,(2020).
- j) **[R1-10]** Two-layer 2D RLC circuits:
Ammar, N, Asad, J, Jarrar, R. "Electrical characteristics for triangular resistors–capacitors–inductors network designed on two coats," Int J Circ Theor Appl., 50(1): 153- 174 (2022).
- k) **[R1-11]** LC network:
Fang, XY., Tan, ZZ. "Circuit network theory of n-horizontal bridge structure," Sci Rep 12, 6158 (2022).

Despite this, nearly all of these papers are theoretical; hence, reporting the experimental measurement of oscillating impedance in this work is important.

Response: We are grateful to the reviewer for recognizing the importance of our work. We believe our manuscript not only contributes to fundamental understanding of the circuit size dependent resonances but also verifies them through the experimental realization. The experimental realizations [1], indeed not many, found in current literature present the two-point impedance in the trivial homogeneous circuits with single type components. Therefore,

we are pleased to present one of the first experimental realizations demonstrating the impedance resonances in LC circuits with the variation of the circuit size. Our experimental results affirm the existence of the size-dependent oscillating impedances and their fractal counterparts through a 7×7 square LC circuit array. It illustrates also that the impedance resonances are resilient against the presence of parasitic and realistic resistances, as well as contact resistances. We therefore believe that this experimental realization is a significant contribution to the field.

We also thank the reviewer for furnishing us with a plethora of research papers pertaining to the two-point impedance behaviors in several LC circuit models. We acknowledge the significance of these papers, which explain, among other things, the impedance characteristics of the unique circuit networks under examination in the presence of resonant components. As such, we have cited the shared papers in the new paragraph in the introduction section colored by red. (R1-1 $\rightarrow$ [Ref. 55], R1-2 $\rightarrow$ [Ref. 58], R1-3 $\rightarrow$ [Ref. 32], R1-4 $\rightarrow$ [Ref. 18], R1-5 $\rightarrow$ [Ref. 43], R1-6 $\rightarrow$ [Ref. 41], R1-7 $\rightarrow$ [Ref. 42], R1-8 $\rightarrow$ [Ref. 43], R1-9 $\rightarrow$ [Ref. 40], R1-10 $\rightarrow$ [Ref. 44], R1-11 $\rightarrow$ [Ref. 57].) Consequently, we removed the word “discovery” throughout the manuscript. Nonetheless, we would like to respectfully elaborate upon the key distinctions between those works and our current study.

First, our manuscript primarily emphasizes the concept of circuit size-dependent impedance resonances, in addition to the established LC resonance conditions. While the 3D impedance graphs in the earlier studies (R1-1 (Fig. 7), R1-2 (Fig. 13-16), R1-3 (Fig. 11), R1-4 (Fig. 6-8), R1-5 (Fig. 6-10), R1-6 (Fig. 5-12), R1-7 (Fig. 9-12), R1-8 (Fig. 6-9), R1-9 (Fig. 8-9), R1-11 (Fig. 9-10)) depict the impedance responses in the $\omega^2 LC - n$ domain, it was neither clearly stated nor evident that the observed impedance resonances are a direct consequence of the network size. In contrast, in our study, we investigate the impedance behavior as the circuit size is incrementally enlarged. Our theoretical and experimental findings indicate that an LC circuit can exhibit resonance characteristics that are contingent on the size of the network (Fig. 2(c-f) of the revised manuscript).

Then, we demonstrate that changing the circuit size leads to fractal-like structures, which has not been previously reported in any of the aforementioned papers. Consequently, we regard our results as a novel contribution to the existing studies that complement the findings presented in the reference papers. We have acknowledged these papers with the proper citations in the highlighted new paragraph in the introduction section.

Last, the analytical impedance formula developed for our 2D LC circuit provides a theoretical interpretation of the size-dependent resonances akin to the Harper equation in condensed matter physics, where a fractal pattern is known as the Hofstadter butterfly arises in the energy levels in the presence of a magnetic flux. Therefore, our study not only reveals the size-dependent resonances and their fractal diagrams, but also offers an understanding of their underlying nature in electric circuits.

- 2) To motivate their study, the authors present a one-parameter scaling theory within which the anomalous impedance scales as $Z \propto N^{2-D}$ for $D \neq 2$ and $Z \propto \log N$ when $D=2$. While this is an intuitive picture, I am concerned about its validity and reliability. This is because, for any LC circuit, independent of dimension, one can evaluate the Laplacian and, subsequently, determine the resonance frequencies responsible for irregular oscillations in the imaginary parts of impedance. Hence, independent of the dimension of these resonant circuits, one should be able to determine their resonance frequencies and observe anomalous impedance. Here are

references that reported similar structure in the impedance in quasi-1D (1xn) ladder LC network and n-step 3D cube LC network.

a) [R1-12] Ladder LC network:

Hai-Xiang Chen et al., "Equivalent complex impedance of n-order RLC network," Phys. Scr. 96 075202 (2021)

b) [R1-13] n-step cube network:

Zhen Tan et al., "Electrical characteristics of the $2 \times n$ and $n^2 \times n$ circuit network" Phys. Scr. 94 055203 (2019).

The reported oscillating impedance in both papers is in dimensions different than two. This undermines the authors' argument regarding the importance of $d=2$ and the connection they drew between their results and other 2D systems in other research fields.

Response: We agree with the reviewer and appreciate their insight in pointing out the potential for readers to interpret that uniform scaling violations arise only in 2D circuits. Yes, our results, including fractal-like scaling and size-dependent impedance resonances that contradict the classical picture in fully reactive or resistive circuits, are also observable in lattices of different dimensions. Accordingly, we have revised the second and last paragraphs in the introduction section and made minor alterations to the first paragraph of the "Violation of logarithmic impedance scaling" subsection. We have removed the focus on the 2D case and instead put forward scaling violations in all dimensions. However, as our experiment was conducted on a 2D sample, we continue to discuss the 2D circuit in the Results section. The changes have been highlighted in red, except for minor adjustments. Below, we provide the notions of our key modifications that we made in the revised manuscript.

Systems with $D = 2$ exhibit a unique behavior in which the conductivity increases logarithmically with the system size [2]. Alongside its presence in various scaling scenarios, this logarithmic scaling can also be seen in electrical circuits when the system size acts as a functional parameter. In our study, we specifically investigate the changes in impedance characteristics of a 2D electrical circuit to demonstrate how this well-established picture can break down even in simple electrical circuits when non-trivial components are present.

Previous studies on electric circuits with different dimensionalities have shown that the impedance or resistance in a fully reactive or resistive circuit saturates to a constant impedance as the size increases when $D > 2$ [3,4,5]. Since the case $D = 2$ presents a unique example where the impedance increases logarithmically, we decided to examine the size dependency of the impedance in the simplest 2D LC circuit even though the violation of uniform impedance increment can also be observed in circuits of all dimensions. With the aim of providing a more comprehensive understanding for the reader, we have included references R1-12 $\rightarrow$ [Ref. 17] and R1-13 $\rightarrow$ [Ref. 18] in the second paragraph of the introduction section of the revised manuscript, offering a broader range of examples for their consideration.

3) Here are other comments and questions.

a) The manuscript should include a proper introduction to previous works on complex impedance networks.

Response: We would like to thank the reviewer for pointing out the absent background in the introduction of our manuscript. We have revised the introduction section to include a comprehensive literature review of existing works on the two-point impedance in complex impedance networks. This additional third paragraph highlighted in red provides an in-depth examination of the prior research in this field and the context and background for the study at hand.

b) For better readability of this work in the physics community and to keep the manuscript self-contained, it would be helpful if the authors present the relation between impedance, resistance, and reactance, either as a footnote or in the text.

Response: In accordance with the reviewer's suggestion, we have included the main definitions and distinctions between the impedance, resistance, and reactance in the second paragraph of the "**Detailed derivation of the corner-to-corner impedance**" subsection of the "**Method**" section (highlighted in red). This provides a clear and concise explanation for readers who may be unfamiliar with these concepts.

c) As Z is imaginary in LC circuits, one should use $|Z|$ instead of Z in the definition of real-valued β , plotted in Fig.1.

Response: We are grateful to the reviewer for bringing the missing modulus to our attention. We have now corrected the errors in the manuscript to ensure the technical accuracy and clarity of our work.

d) For the benefit of readers, the recursive relation of the circuit should be calculated and added in the main text, which can then be directly compared with Harper's equation.

Response: Thank you for your insightful comment regarding the recursive relation of the circuit. We appreciate the potential value of including it in the main text to provide readers with a direct comparison to Harper's equation. However, we have opted not to include the recursive relation in our manuscript for the following reasons.

While we believe the Recursion-Transform (RT) method introduced by Tan [6,7,8] is one of the most useful methods for calculating the two-point impedance in finite and infinite lattices, this method involves several steps, including constructing a recursive matrix, solving it, and applying specific boundary conditions. Although successful in calculating the equivalent complex impedance in LC circuits, deriving the recursive relation using the RT method requires several pages of detailed presentation, as demonstrated in references [6,7,8].

Our manuscript primarily focuses on presenting our experimental results. Including a second theoretical calculation method, such as the recursive relation, alongside our analytical method for calculating the corner-to-corner impedance could potentially confuse readers. We have already mentioned the method in the third paragraph of the introduction section and stated in the first paragraph of the "**Detailed derivation of the corner-to-corner impedance**" section that readers can compute the two-point impedance using either the RT or method of images approaches, according to their preference. Additionally, we have provided relevant reference papers for further guidance [Refs. 59,60,91].

Despite our choice not to include the recursive relation, we acknowledge its usefulness in calculating the two-point impedance. If the reviewer strongly believes that incorporating the recursive relation would significantly benefit the manuscript, we are more than willing to do so.

- e) The authors described the anomalous structure of impedance as fractal-like. Could they determine the fractal dimension or the rescaling relation? For resistance networks, a simple rescaling ratio, not fractal though, in certain geometries is introduced here: Rajat Chandra Mishra and Himadri Barman, “Effective resistances of two-dimensional resistor networks” Eur. J. Phys. 42 015205 (2020).

Response: In the revised manuscript, we have added an additional subsection titled “**Determination of the Fractal Dimension**” under the “**Method**” section. There, to estimate the fractal dimension, we used singular value decomposition (SVD) [10,11,12,13]. We studied Fig. 3a in our manuscript and determined its scaling ratio as ~ 0.7 and its fractal dimension as $D = 1.4$. As presented in [14,15,16,17,18], the fractal dimension can be found by determining the slope of the log-log scale plot of the singular values obtained from a dataset. We decomposed the $\text{Log}|Z|$ dataset of Fig. 3a and plotted the singular values of $\text{Log}|Z|$ in Fig. 9 in the revised manuscript. The fractal dimension and the scaling ratio, found to be 1.4 and 0.7 respectively, appear reasonable since the fractal dimension is expected to be inverse of the scaling ratio, i.e., $1/0.7 \sim 1.4$.

The paper mentioned by the reviewer provides a proportionality constant that demonstrates the scaling characteristics of various lattice models when the circuit size is altered in either vertical or horizontal directions. Due to the relevance of the scaling discussion presented in this paper, we have cited it in our new subsection “**Determination of the fractal dimension**” with the reference number [Ref. 100].

- f) The authors also discussed the fractal parameter space profile of the impedance. In disordered conductors, one may encounter multifractality; see, for instance, Phys. Rev. Lett. 94, 244102 (2005) and Phys. Rev. E 73, 036210 (2006). In their study, could authors draw or rule out a link between multifractality in conductors and their reported fractal structures?

Response: We appreciate the reviewer’s contribution to furthering the fractal discussion in our context. The multifractality in the shared papers describes the electrical conductivity of disordered conductors based on scaling exponents. These exponents determine diffusive and localized behaviour as the system size increases. According to the Anderson localization interpretation, the scaling exponent is the system dimension ($D = 2$ in our case), which is constant in the studied circuit. And, importantly, our fractal structure is due to the commensurability of the resonant modes with respect to the circuit lattice, which is physically distinct from the mechanism behind multifractality in disordered conductors.

As a result, a connection between the multifractality and our fractal diagrams cannot be made. Nevertheless, to provide more information about multifractality, we have included relevant references in the subsection “**Violation of logarithmic impedance scaling**” with the reference numbers [Refs. 77,78] in our revised manuscript.

Response to Reviewer 2

Main Comment

In this manuscript, the authors find an anomalous impedance with pronounced and erratically located peaks in a 2D LC lattice, which is different from the logarithmic impedance scaling in the continuum media. Such anomalous fractal scaling of impedance is confirmed in experiments, and attributed to the circuit resonance, which is related with commensurability conditions for the vanishing of circuit Laplacian eigenvalues. They further show that such impedance peaks are robust against the contact and parasitic resistances. I think this research is timely and interesting and, to the best of my knowledge, scientifically sound, thus recommend it for publication in Communications Physics.

Response: We greatly appreciate the reviewer's recognition of the novelty and significance of our manuscript, as well as their recommendation for publication in Communications Physics. Indeed, in our manuscript, we present anomalous impedance responses that arise from the variation of the circuit size rather than the circuit parameters. While this property has been observed in various papers [for example, R1-1, R1-2,...,R1-11], the generic conditions and fractal nature have not been discussed extensively. Most importantly, our manuscript not only provides a comprehensive discussion but also demonstrates the existence of these properties through the experimental implementation of a 2D LC circuit.

We have carefully considered the questions and recommendations provided by the reviewer, and we present our responses below.

Comments and questions

1. Why do the impedance peaks in experiments shift to right regularly? It may be not because of the uncertainly tolerances of the electronic elements. Maybe it is owing to the global loss induced by the inevitable resistances. I suggest the authors can give a more detail discussion about this point.

Response: We thank the reviewer for their question and for highlighting the impact of inevitable resistances. Accordingly, we have added a new section titled “**Effects of Inevitable Parasitic Resistances**” to the Method section. As the reviewer stated, shifts in impedance indeed result from both inevitable resistances and uncertainty tolerances. In the new section, we discuss how inevitable parasitic resistances cause a shift in the fractal domain. To investigate this, we considered two cases in which the parasitic resistances R_C and R_L are relatively small and large. From the perspective of the admittance band structure, the admittance band retains its general structure even when we set very large parasitic resistances (e.g., $R_C = R_L = 0.2\Omega$), as shown in Fig. 8b and 8d in the main text. From the numerical Laplacian viewpoint, impedance resonances arise due to the presence of very small eigenvalues. As long as the admittance bands cross the zero-admittance axis, a strong impedance resonance occurs. As depicted in Fig. 8 of the revised manuscript, the effect of parasitic resistances is solely on the band crossing point, i.e., ω . We can thus attribute the

smooth shift in the fractal diagrams to the inevitable resistances along with the uncertainty tolerances and have provided additional details in the new subsection.

2. Why is the range of w_r in Fig. 3(a) and 3(c) different? Also the color used to denote the value of $\log(Z)$? The same range and color would be more easily to compare these two results, revealing the effect of the inevitable resistances.

Response: We are grateful to the reviewer for their suggestion and agree with them. We have addressed the issue in the figures by incorporating the same ranges and ensuring consistency in the use of color palettes in Fig. 3 of the revised manuscript.

3. Although the authors write a sentence to discuss the case of generic circuit, I hope they can discuss the case of honeycomb lattice of LC with 2 site in a unit cell in detail, which should be an important extension of this work.

Response: We thank the reviewer for the chance to elaborate on this. Yes, we have thoroughly evaluated our findings across various lattice models and dimensions, however, we have chosen to present our results in the context of a simple two-dimensional LC circuit for simplicity and clarity in the manuscript. In light of the reviewer's interest, we have also conducted our analysis on a honeycomb lattice and present the results in the revised manuscript and below.

We present the variation of the impedance response and resonances of a honeycomb lattice with the lattice size. As in the manuscript, we measure the impedance between the two furthest corner nodes as the circuit size increases. Fig. RL3 shows the studied lattice model comprising two sublattice nodes. When the node links (depicted by blue and red lines) have opposite phases, the circuit becomes an LC resonant circuit and exhibits size-dependent resonances in addition to the conventional LC resonances. Therefore, the honeycomb circuit serves as a validation of the key results presented in our study.

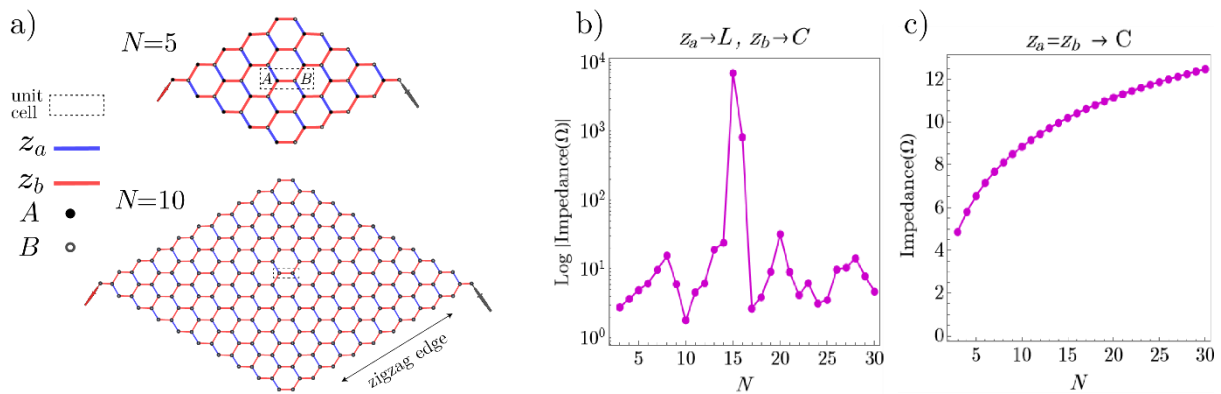


Figure RL1: (a) The lattice structure of the studied honeycomb circuit with zigzag edges. We measure the impedance between the two furthest nodes as the circuit size increases. The blue and red lines between nodes represent the node connections with different admittances z_a and z_b , respectively. The circuit is resonant when z_a and z_b have the opposite phases, i.e., one acting as an inductor and the other as a capacitor. (b), (c) The impedance response of the honeycomb circuit as a function of the circuit size N . The parameters are (a) $\omega C = 1, \omega L = 2.21$ and (b) $\omega C = 1$.

In Fig. RL3b, we observe sharp impedance resonances as a function of the circuit size in the presence of both capacitors and inductors. When the circuit is a uniform homogeneous

system in which all the links are of a single type, the impedance response is logarithmic in nature and increases uniformly. Moreover, we plot the circuit size-frequency diagram in Fig. RL4, from which fractal-like patterns similar to those in our 2D LC circuit can be clearly seen, indicating the generality of the results presented in the manuscript.

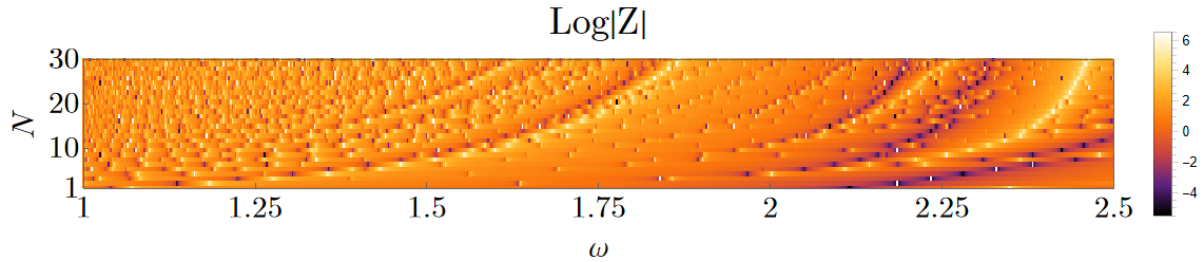


Figure RL2: Fractal diagram of the LC honeycomb circuit with a narrower scale, as compared to Fig. 4c in the revised manuscript, allowing for more details to be seen. There are strong impedance resonance fractal branches as the circuit size N increases. This result validates the universal character of the fractal-like properties of the impedance in the circuit size frequency domains. The resolution of the branches can be enhanced by running the computation longer with smaller steps. The parameters are $C = L = 1$.

4. I suggest the authors can optimize the distribution of the figures, to make them look clear. For example, six figures in a row in Fig. 2 result in each figure small. And the labels of figures, (a),(b),(c)...., are better to locate at the left top of the figures.

Response: We have meticulously reorganized the layout of the plots in Fig. 2 for improved clarity and coherence. The newly updated figure provides a clearer representation, ensuring that more intricate details can be easily discerned and comprehended by the reader. In addition, we have refined all figures in the manuscript for enhanced visualization and presentation.

Response to Reviewer 3

Main comment

In their paper, Xiao Zhang and coauthors consider a circuit in the form of the square lattice with horizontal and vertical links between adjacent nodes being capacitors and inductance coils, respectively. The authors reveal counterintuitive dependence of the corner-to-corner impedance on the number of nodes in the circuit, which demonstrates the emergence of pronounced peaks for certain numbers of nodes (considering some given frequency of the excitation) instead of a logarithmic scaling. In my opinion, the paper needs to be revised prior publication.

Response: We sincerely appreciate the reviewer's valuable comments and suggestions on our manuscript. Following their guidance, we have carefully revised our manuscript to more clearly and effectively convey our findings. We believe that the reviewer's suggestions have significantly enhanced the quality of our manuscript. Our responses to each of the reviewer's questions are provided below.

Comments and questions

1. First, the authors include a lot of references to papers considering topological electrical circuits (most of them being self-citations) many of which can be omitted as the present paper is not related to topological effects, while not including any citation for the considered circuit design. The studied circuit is not novel and has been considered, for example, in [Z. Tan, Chin. Phys. B 25, 050504 (2016)], where the expressions for impedance between two arbitrary nodes at the same side of the circuit were obtained. This reference should be included in the manuscript, and I would like to recommend the authors to check carefully if the considered circuit has been studied in any other papers. Also, slightly modified circuits were considered to implement hyperbolic metamaterials [P. Kapitanova, et al., Nat. Commun. 5, 3226 (2014)].

Response: We thank the reviewer for pointing out the missing references. We acknowledge that the circuit in our study has been extensively researched in the field of circuit theory. Considering this, we have conducted a thorough literature review and have included the relevant references in a new paragraph of the introduction section (highlighted in red) to provide the necessary context for the readers.

As the reviewer raised in their next question, the unique aspects of our work are the previously undeclared circuit size-dependent impedance resonances, their fractal-like diagrams, and their commensurability properties. These features are generic and can be found in all lattice models, provided that at least two components with opposite phases are present. We therefore believe that the findings of our study are applicable not only to the 2D square circuit but also to all LC circuits in general, representing significant contributions to the field.

Regarding the references, we have removed many of our self-citations and included the two suggested references in the third paragraph of our revised manuscript [Refs. 55,77], due to their relevance to our work. The study in the reference by Z. Tan presents a unique method, known as the recursion transform, for calculating the two-point impedance behavior in finite LC circuits. Their study offers insights into the fractal nature of strong impedance resonances. We find Z. Tan's study to be significant and have therefore cited their work in our revised manuscript. The second reference, by P. Kapitanova et al., demonstrates that even simple circuits can be employed to design metamaterials for examining transmission phenomena at lower frequencies. For instance, their study reveals that a 2D transmission line design with capacitors and inductors uniformly aligned along both directions can mimic a hyperbolic metamaterial that exhibits unidirectional propagation in accordance with the polarization of the excitation. We have also incorporated P. Kapitanova et al.'s work into our revised manuscript and would like to take this opportunity to thank the reviewer for bringing these references to our attention.

2. I think that the novelty of the present manuscript is not in the circuit design, but in revealing commensurability properties and establishing a connection between such a circuit and properties of electrons in a magnetic field, which seem to be a very interesting result. In the light of such an analogy, I would also like to attract the attention of the authors to the paper [A.V. Poshakinskiy, et al., npj Quantum Inf. 7, 34 (2021)] considering a correspondence between Aubry-Andre-Harper model and the system of coupled microwave qubits located in a waveguide.

Response: We thank the reviewer for pointing out one of the novelties in our work. Yes, our work establishes an analogy between the impedance scaling in electric circuits and the spectral properties of electrons in a magnetic field. We also find that the shared paper is relevant to our analogy and provides valuable context for our research. In addition to the previous question/comment by the reviewer, we would like to elaborate on the size-dependent impedance resonances and their fractal-like patterns in the size-parameter domain. While fractal spectra are observed in a variety of systems, their commensurability is often quite limited and challenging to achieve. For example, as shown in the mentioned paper, a synthetic magnetic field can be created through atom-photon interactions in an array of two-level atoms coupled to a waveguide, leading to the emergence of topological order in a relatively simple quantum structure. This results in the emergence of quantum Hall phases, topological edge states, spectral Landau levels, and Hofstadter butterfly spectra. Considering the valuable contribution of this work to our discussion, we have cited the paper in the paragraph below Eq. 3 in “**Emergent fractal resonances**” section with the reference number [Ref. 89] in our revised manuscript and would like to express our appreciation to the reviewer for bringing it to our attention.

3. Regarding the discretization of continuous media and its representation with electrical circuits, I would like the authors to consider mentioning well known papers [S. Kirkpatrick, Rev. Mod. Phys. 45, 574 (1973); I. Webman, et al., Phys. Rev. B 15, 5712 (1977)]. Besides, some references on more general networks should be introduced in Discussion to illustrate the statements written in this section.

Response: We strongly agree with the reviewer regarding the significance of the two shared papers, and we have included both references in a new statement on the importance of discretization in the Discussion section. The paper by S. Kirkpatrick serves as one of the foundational works on resistor networks, providing a generic framework for the conductivity problem of lattice models. The second paper, by Webman et al., employs the discretization method to present a theory for light propagation in heterogeneous media. Their numerical approach, as outlined in their paper, solves the Laplace equation for complex dielectric constants in a cubic network by discretizing the lattice into small cubes. This enables them to study the scattering and conductivity properties of the medium in the context of light propagation. We find both papers highly relevant to our study; indeed, they have paved the way for our research. As such, we have cited the two papers shared by the reviewer with reference numbers [Refs. 20, 90].

4. Also, there are several minor concerns that should be addressed in order to improve the general quality of the paper.

- a. There are several typos and mistakes in English throughout the text.

Response: We have carefully rechecked our manuscript for language errors and made the necessary corrections.

- b. Many of the references contain various imperfections in formatting.

Response: We have updated our references and fixed the formatting imperfections.

- c. In Introduction and other sections of the manuscript, LC circuits should be mentioned as a resonant media instead of just a reactive one, as purely capacitive or inductive circuits are reactive as well. However, they obviously will not demonstrate any unusual behavior on the number of nodes in the circuit.

Response: We are grateful to the reviewer for bringing to our attention the possible misinterpretation of the statements concerning the LC circuits. We have made the necessary corrections to such statements throughout the manuscript to ensure clarity and accuracy. The changes are highlighted in red.

- d. The paper is clearly organized and well written.

Response: We would like to express our appreciation to the reviewer for the kind comment. We have taken great care to maintain the overall structure and organization of the submitted version of our manuscript while making various changes in response to the reviewers' suggestions to ensure coherence and a clear and logical flow of information, which are important for the reader's understanding.

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REVIEWERS' COMMENTS:

Reviewer #1 (Remarks to the Author):

I thank the authors for their responses to my questions and for implementing most suggestions in the manuscript.

As I have pointed out in my previous report, and the authors accepted, fractal behavior is realizable in all dimensions. With respect to this and to avoid further confusion for readers, the following line of the manuscript and the legend of Fig.1(b) should be improved to become more precise.

"As plotted in Fig. 1b (red) for an illustrative AC circuit lattice (detailed later), β fluctuates erratically and dramatically as the system size N increases, some- times even exceeding the constant β of other dimensionalities (dashed)."

The reason for this revision is that the constant value of β in Fig. 1(b) is obtained from $\beta = D - 2$, which should not be confused with the fractal behavior of impedance in LC circuits in $D \in \{1, 3\}$.

The only unresolved issue is the possibility of extension of the introduced analogy between the impedance scaling in (R)LC circuits and the behavior of electrons in a magnetic field in dimensions other than two. For this concern, I suggest authors consider mentioning papers such as [M. Koshino et al. Phys. Rev. Lett. 86, 1062 (2001)] where Harper's equation for 3D, effectively quasi-1D, systems are discussed.

By implementing these two minor modifications, the manuscript should be improved considerably and will be at the stage to be recommended for publication.

Reviewer #2 (Remarks to the Author):

The authors did a good job in revising their manuscript. All my previous questions have been addressed properly. I believe the paper can be accepted for publication in Communications Physics.

Reviewer #3 (Remarks to the Author):

The authors addressed all my main recommendations, and considerably extended the manuscript following the questions by other Referees. I ask the authors to check the text for typos again, as I still have found some.

In my opinion, the manuscript in its current form meets the high level of Communications Physics and can be accepted.

Response to Reviewers' Comments/Questions

Response to Reviewer 1

Comments and questions

1. As I have pointed out in my previous report, and the authors accepted, fractal behavior is realizable in all dimensions. With respect to this and to avoid further confusion for readers, the following line of the manuscript and the legend of Fig.1(b) should be improved to become more precise. "As plotted in Fig. 1b (red) for an illustrative AC circuit lattice (detailed later), β fluctuates erratically and dramatically as the system size N increases, some- times even exceeding the constant β of other dimensionalities (dashed)." The reason for this revision is that the constant value of β in Fig. 1(b) is obtained from $\beta=D-2$, which should not be confused with the fractal behavior of impedance in LC circuits in $D \in \{1,3\}$.

Response: We are grateful for the previous review by the reviewer and appreciate their further suggestions for improving the quality of our study. In response to the reviewer's additional suggestions, we have removed the statement "sometimes even exceeding the constant β of other dimensionalities (dashed)" from the second paragraph of the "Violation of logarithmic impedance scaling" subsection. Instead, we have included a new statement (highlighted in red) to ensure that readers are aware of the multi-dimensionality of the fractal scaling. Additionally, we have revised the legends of Fig. 1b and the corresponding caption to de-emphasize the case of 2D scaling.

2. The only unresolved issue is the possibility of extension of the introduced analogy between the impedance scaling in (R)LC circuits and the behavior of electrons in a magnetic field in dimensions other than two. For this concern, I suggest authors consider mentioning papers such as [M. Koshino et al. Phys. Rev. Lett. 86, 1062 (2001)] where Harper's equation for 3D, effectively quasi-1D, systems are discussed.

Response: We have incorporated a statement (highlighted in red) in the first paragraph of the "Emergent fractal resonances" subsection and included the reference Phys. Rev. B 65, 045310 in addition to the suggested reference Phys. Rev. Lett. 86, 1062. We strongly believe that the reviewer's suggestions have successfully eliminated any ambiguous phrases regarding the occurrence of erratic scaling in systems of all dimensions, rather than being limited to only two-dimensional systems.

Response to Reviewer 2

The authors did a good job in revising their manuscript. All my previous questions have been addressed properly. I believe the paper can be accepted for publication in Communications Physics.

Response: We appreciate your positive evaluation and the time you have invested in reviewing our manuscript. Your insights have greatly contributed to the improvements we have made. We are pleased to hear that our responses addressed your earlier queries satisfactorily and we are encouraged by your endorsement for publication in Communications Physics. Thank you once again for your valuable contributions.

Response to Reviewer 3

The authors addressed all my main recommendations, and considerably extended the manuscript following the questions by other Referees. I ask the authors to check the text for typos again, as I still have found some.

In my opinion, the manuscript in its current form meets the high level of Communications Physics and can be accepted.

Response: We are very thankful for your constructive feedback and positive assessment of our manuscript. We also apologize for the typos and mistakes in language. Before we publish, all authors will double-check the text in proofreading to make sure there are no errors.

Your input has contributed the improvement of our manuscript and clarified the missing points. Once again, thank you for your valuable insights and your time spent reviewing our work. We sincerely hope that our research will contribute meaningfully to the scientific community.